



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Parallel Computing– II part– Thursday, July 04th, 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how the parallel split pattern works and is parallelly implemented. (2)

[illegible]

B. Please describe what **relaxing** the sequential consistency memory model means and why it is done. (2)

[illegible]

Exercise n. 2

Answer the following questions about OpenMP (without an explanation, the answer will be considered invalid).

A. Describe in detail the effect of the following OpenMP pragma. (2)

```
#pragma omp parallel for shared(tmp) private(i, j) if (!embedded)
```

B. Which OpenMP pragma allows offloading part of the computation to an accelerator? Which clause is used to manage data transfers? (1)

C. List the factors that determine the number of threads in an OpenMP parallel region (1)

Exercise n. 3

Answer the following questions about CUDA (without an explanation, the answer will be considered invalid).

- A. You are tasked with accelerating on GPU an application that contains a) pre-processing of images implemented as a set of for loops; b) a deep neural network; c) control-intensive post-processing of the results. Which programming model would you choose to implement each of the three components? Motivate your answers. (3)

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- B. Control divergence can affect the performance of a CUDA program containing if-else blocks. Is this statement true or false? Why? (1) **This question can be skipped in case of a passed challenge.**

[illegible]

Exercise n. 4

Answer the following questions about parallel programming (without an explanation, the answer will be considered invalid).

A. Why is synchronization necessary when programming with the shared memory model? (1)

B. Sometimes a parallel program can be slower than a sequential one. Is this statement true or false? Why? (1)

C. What is a DSL? (2)
